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NGN: Learning Neural Network Size as a Differentiable Count

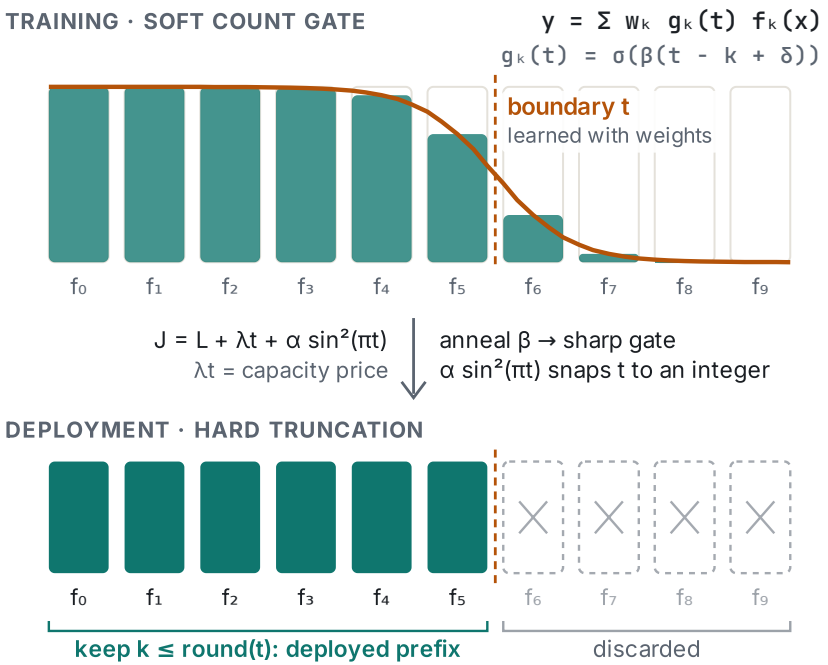
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1 PROBLEM

- Widths, depths, and other structural sizes are usually **fixed before training**; choosing them takes expertise or repeated training across candidate sizes.
- Useful capacity depends on the data and objective, so size could instead be **optimized alongside the weights**.
- NGN learns **how many** ordered components a model needs as one differentiable boundary per component group.

MLP · CNN · GNN · Transformer · Mamba · LoRA · Adapters · Depth · Structured pruning

2 METHOD



One learnable boundary t gates an ordered prefix of components f_k (scaled by $w_k = 2^{-k}$). The capacity price λt pushes t inward; task gradients at the frontier recruit new components.

3 KEY RESULTS

49 / 50

seed–dimension cases where the learned VAE bottleneck count matches a nonlinear manifold's true intrinsic dimension (98%)

0.00 pt change

accuracy change from hard truncation on all four 10-seed MNIST, CIFAR-10, and Cora rows (medians; MNIST MLP 96.93 → 96.93)

78.6 vs 75.2 / 45.1

WikiSQL-easy score of truncated NGN LoRA on Llama-3-8B-Instruct at learned rank 19, vs SoRA / AdaLoRA at the same rank